

OBJECTIVE

Specialising in network automation and data center architecture building spine-leaf fabrics, SDN controllers, and automated lab environments from scratch. Goal: work at the intersection of infrastructure and intelligence, designing networks that are faster, smarter, and more resilient, contributing to systems that power businesses, connect communities, and move society forward. Actively pursuing CCNA certification and seeking a team where I can grow fast and help build something that matters.

TECHNICAL SKILLS

Protocols	BGP (eBGP/iBGP), OSPF, VXLAN, EVPN, OpenFlow, ECMP, STP
Automation	Ansible, Containerlab, FRRouting (FRR), Infrastructure as Code (YAML)
SDN / Virt	Mininet, Open vSwitch, Faucet Controller, Docker, Docker Compose
Observability	Prometheus, Grafana, cAdvisor
Programming	Python (network scripting & automation), Bash / Shell
OS / Tools	Kali Linux, Ubuntu, Git, GitHub, VS Code

PROJECTS

► NetMind AI-Powered Network Operations Assistant

github.com/MrWater00/netmind

Python · FastAPI · React · Groq API (Llama 3.3 70B) · LLM Tool Use

Built an autonomous AI agent that diagnoses network faults in plain English ask "why is my BGP session down?" and it runs the right commands, correlates findings, and returns a root cause with fix.

Implemented LLM function calling loop where the AI autonomously decides which diagnostic commands to run across a 5-router simulated topology with real BGP, OSPF, and interface faults.

► AWS Network Automation

github.com/MrWater00/aws-network-automation

Terraform · Python · boto3 · LocalStack · Docker

Automated provisioning of a complete AWS VPC network stack public/private subnets, internet gateway, security groups, and route tables entirely through Terraform IaC and Python boto3 scripts.

Developed and tested entirely against LocalStack, a local AWS emulator, enabling full cloud network automation without live AWS credentials or cost.

► Network Lab Orchestrator

github.com/MrWater00/network-lab-orchestrator

Python · Ansible · Containerlab · Prometheus · Grafana · Docker

Automated IaC platform deploying, configuring & monitoring full network topologies from a single YAML file via Ansible + FRRouting.

Integrated Prometheus + Grafana + cAdvisor observability stack for real-time CPU, memory & network I/O dashboards across all lab containers.

► Spine-Leaf Data Center Fabric

github.com/MrWater00/spine-leaf-fabric

Shell · Containerlab · FRRouting · eBGP · VXLAN · BGP EVPN

Production-grade 2-spine/2-leaf Clos fabric (same architecture as AWS & Google Cloud) with eBGP underlay, ECMP load balancing, and VXLAN/BGP EVPN overlay.

Eliminated MAC flooding and achieved clean L2 extension across L3 fabric using BGP EVPN control plane.

► SDN Load Balancer

github.com/MrWater00/sdn-load-balancer

Python · Mininet · Open vSwitch · Faucet · OpenFlow 1.3

Round-robin load balancer distributing HTTP traffic from a single VIP across 3 backends using programmatic OpenFlow rules with full IP/MAC rewrite at switch level.

Debugged Python 3.13 compatibility migrated from Ryu to Faucet, resolved all flow rule conflicts and ARP challenges independently.

CERTIFICATIONS & EDUCATION

Cisco CCNA · Self-studying · In Progress

BGP, OSPF, VLANs, STP, ACLs, NAT, network security reinforced daily with live Containerlab labs.

BCA Data Science & AI

SRMU, Lucknow · Completed May 2026 · Result Awaited

ADDITIONAL WORK

dns-labs, bgp-labs, shadowScan (Go), gopherShell (Go), HoneyPot (Python), ai-chatbot & more →

github.com/MrWater00